

Department of Mathematics
Dhaka College, Dhaka
3rd year honors Examination-2023
Math Lab-II, FORTRAN (Assignment Questions)

1. A function $f(x)$ is defined by $f(x) = \begin{cases} -x & \text{if } x \leq 0 \\ x & \text{if } 0 < x < 1 \\ 1-x & \text{if } x \geq 1 \end{cases}$. Write a FORTRAN program segment to evaluate above function.

2. A function is defined as follows: $f(x) = \begin{cases} 1-x^2, & x < 0 \\ 1+2x, & 0 \leq x < 1 \\ 3+\frac{1}{x}, & x > 1 \end{cases}$ If $x \in [-10, 10]$ then find the value of $f(x)$ by using FORTRAN for each $x = 0.5$

3. Consider $f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2}, & \text{When } (x, y) \neq (0, 0) \\ 0, & \text{When } (x, y) = (0, 0) \end{cases}$ Write a FORTRAN program which find $f_{xy}(0, 0)$ and $f_{yx}(0, 0)$ and print $f_{xy}(0, 0) \neq f_{yx}(0, 0)$.

4. Compute $\int_0^6 \frac{dx}{1+x^2}$ by Simpson's $\frac{1}{3}$ rule and Simpson's $\frac{3}{8}$ rule and compare your result to exact value.

5. Find the dominant eigenvalue and the corresponding eigenvector of the following matrix: $\begin{pmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ with $X^{(0)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

6. Compute $\int_0^{\frac{\pi}{2}} e^{\sin x} dx$ corrected up to 5 decimal places by Simpson's $\frac{3}{8}$ rule and Trapezoidal rule. After finding the true value of the integral, compare the errors in these cases.

7. Write a FORTRAN program to find the sum of z , where $z = x^2 - 2xy + 3y^2 - 8x + 3y - 8$ for the values of x and y range -3 to 3 with increment 0.2.

8. Solve the following system correct to five decimal places

$$2x + 4y + 6z = 22$$

$$3x + 8y + 5z = 27$$

$$-x + y + 2z = 2$$

by using LU factorization method. Also show L and U.

9. Solve the following system correct to five decimal places

$$2x + 4y + 6z = 22$$

$$3x + 8y + 5z = 27$$

$$-x + y + 2z = 2$$

by using SOR iterative method with three different values of ω .

10. For Romberg integration to approximate the value of the integral $\int_0^{\pi/4} x^2 \sin x dx$ correct up to five decimal places. Compare your result with the exact value 0.0887553. Show your result in a tabular form.

11. Write a FORTRAN program, which prints the first 20 Fibonacci numbers and determine whether each number is prime or not.

12. Write a FORTRAN program to find the product $AB = C$ where A and B are defined as

$$A = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 5 & 7 \\ 3 & 8 & 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 & 2 & 15 \\ 2 & 3 & 7 \\ 13 & 8 & 10 \end{pmatrix}.$$

13. Use LU decomposition method to solve the system of linear equations: $AX = B$ where

$$A = \begin{pmatrix} 4 & 1 & -1 \\ 2 & 7 & 1 \\ 1 & -3 & 12 \end{pmatrix}, X = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 \\ 19 \\ 31 \end{pmatrix}. \text{ Show the results of L and U and solutions.}$$

14. the following series:

$$(i) \left(1 + \frac{1}{2}\right) + \left(1 + \frac{1}{2} + \frac{1}{3}\right) + \dots + \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{100}\right) \quad (ii) 1 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{25!}; \text{ where } n \in N$$

15. Write a FORTRAN program to find it output, to obtain the root of the equation $\cos x = 3x - 1$ correct to four decimal places, using fixed point iteration method.

16. Write a FORTRAN program for bisection method to find a root of the equation $e^{-x} - x = 0$

17. Write a FORTRAN program to find the largest Eigen value of the matrix $\begin{pmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{pmatrix}$

18. Use Newton's forward difference interpolation formula to evaluate $F(2003)$ for the following set of tabulated values:

X	2000	2005	2010	2015	2020
Y=F(X)	100	125	139	200	250

19. Write a FORTRAN program which reads a 5×5 matrix and find its inverse matrix.

20. Write a FORTRAN program to find the product of AB, Where A and B are matrices of order $l \times m$ and $m \times n$ respectively.

21. Using Lagrange's interpolation formula of 2-points, 3-points and 4-points, write a program to estimate $f(6.0)$ employing the following data sets

x	1.0	4.0	9.0	16.0
f(x)	101.0	102.0	103.0	104.0

22. A class of 20 students takes the four test on computer course. Write a FORTRAN program which finds

- (i) The average score
- (ii) The no. of first class scores (i.e. score $\geq 60\%$)
- (iii) The no. of second class scores (i.e. score $45\% \leq \text{score} \leq 60\%$)
- (iv) The no. of third class scores (i.e. score $40\% \leq \text{score} \leq 45\%$)
- (v) The no. of those which failed.

23. Write a FORTRAN program to solve $\frac{dy}{dx} = x + y^2$, $y(0) = 1$ by using Picard's method.

24. Write a FORTRAN program to compute $y(0.2)$ by Runge-Kutta method of order for the differential equation: $\frac{dy}{dx} = xy + y^2$, $y(0) = 1$.

25. Write a FORTRAN program to find the roots of the equation $\cos x - xe^x = 0$ up to 4-decimal places by using Newton-Rapson's method.

26. Write a FORTRAN program to minimize the objective function

$$\text{Max } [15x + 18y]$$

$$\text{s.t } 5x + 3y \geq 2$$

$$3x + 6y \geq 3$$

$$x, y \geq 0$$

27. Write a FORTRAN program to solve the system following system of equations using Gauss-Seidal method. $27x + 6y + z = 85$; $6x + 15y + 2z = 72$; $x + y + 54z = 110$

28. Consider the differential equation $\frac{dy}{dx} = x^2 + 1$; $y(0) = 0.5$. Write a FORTRAN program to find the solution of the equation for $x = 0$ to $x = 2$ by Euler and modified Euler methods.